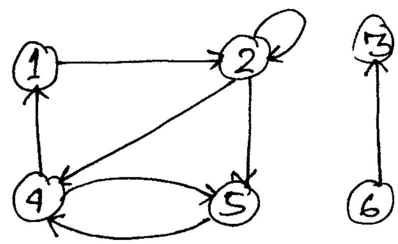
Appendix B.4 : Graphs

1) Graphs are of two types: directed and undirected. A directed graph (or digraph) G is a pair (V,E), where V is the set of vertices and E is the set of edges. In a digraph, self-loops, that is edges from a vertex to itself, are possible.



→ In an undirected graph G = (V,E), the edge set E consists of an unordered pair of vertices. In a directed graph (u,v) represents an ordered pair symbolising and edge from u to v. In an undirected graph (u,v) represents an unordered pair, and (u,v) and (v,u) are the same edge. In an undirected graph, self-loops are forbidden, so that every edge consists of exactly two distinct vertices.

2) If (u,v) is an edge in a directed graph, we say (u,v) is incident from or leaves vertex u and is incident to or enters vertex v.

If (u,v) is an edge in an undirected graph, we say that (u,v) is incident on vertices u and v.

3) If (u,v) is an edge in a directed graph, we say v is adjacent to u, which is sometimes written as $u \rightarrow v$. This adjacency relation is not symmetric.

If (u,v) is an edge in an undirected graph, then both u is adjacent to v and v is adjacent to u.

4) Degree of a Vertex:

The degree of a vertex in an undirected graph is the number of edges incident on it. A vertex with degree 0 is said to be isolated.

In a directed graph, the out-degree of a vertex is the no. of edges leaving it. The in-degree of a vertex is the no. of edges entering the vertex. The degree of a vertex in a directed graph is its in-degree plus its out-degree.

5) Path:

A path of length K from a vertex u to a vertex u' in a graph $G = (V, E)$ is a sequence $\langle v_0, v_1, v_2, \dots, v_K \rangle$ of vertices such that $v_0 = u$, $v_K = u'$ and every $(v_{i-1}, v_i) \in E$ for all $i = 1$ to K. Length of a path is the no. of edges in the path. There is always a 0-length path from u to u.

If there is a path p from u to u', we say that u' is reachable from u via p. If G is directed this is written as $u \overset{p}{\leadsto} u'$.

A path is said to be simple if all vertices in the path are distinct.

6) Subpath:

A subpath of path $p = \langle v_0, v_1, v_2 \dots v_K \rangle$ is a contiguous subsequence of its vertices, $\langle v_i, v_{i+1}, v_{i+2}, \ldots, v_j \rangle$ such that $0 \le i \le j \le k$.

7) Cycle:

In a directed graph, a path $\langle v_0, v_1, v_2 \ldots v_k \rangle$ forms a cycle if $v_0 = v_k$ and the path contains at least one edge. The cycle is said to be simple if $v_1, \ldots, v_k$ are distinct. A self-loop is a cycle of length 1.

Two paths $\langle v_0, v_1, v_2, \ldots, v_{k-1}, v_0 \rangle$ and $\langle v_0', v_1', v_2' \ldots v_{k-1}', v_0' \rangle$ form the same cycle if there exists an integer j such that $v_i' = v_{(i+j) \bmod k}$ for $i = 0, 1, \ldots k-1$.

A directed graph that has no self-loops is said to be simple.

In an undirected graph, a path $\langle v_0, v_1, v_2 \ldots, v_k \rangle$ forms a (simple) cycle if $v_0 = v_k$ and $k \geq 3$ and $v_1, v_2, \ldots v_k$ are distinct.

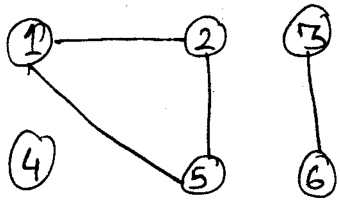
A graph with no cycles is said to be acyclic.

8) Connectedness:

An undirected graph is connected if every pair of vertices is connected by a path.

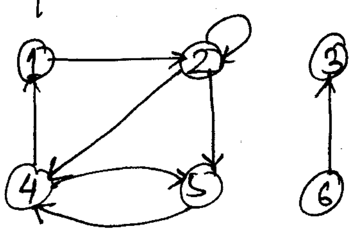
The connected components of a graph are the equivalence classes of vertices under the "is reachable from" relation.

An undirected graph is connected if it has exactly one connected component, that is, every vertex is reachable from every other vertex.



This graph has three connected components: {1,2,5}, {3,6} and {4}.

A directed graph is strongly connected if every two vertices are reachable from each other. The strongly connected components of a directed graph are the equivalence classes of vertices under the "are mutually reachable" relation. A directed graph is strongly connected if it has only one strongly-connected component.

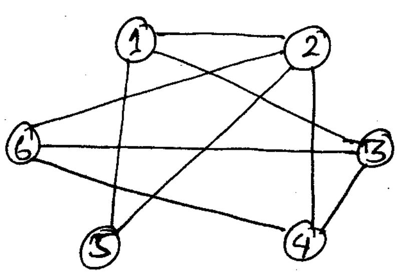


This graph has three strongly connected components: {1,2,4,5}, {3} and {6}. All pairs of vertices in {1,2,4,5} are mutually reachable.

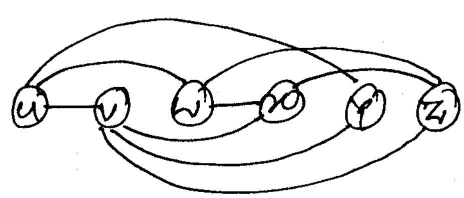
9) Isomorphic graphs:

Two graphs $G=(V,E)$ and $G'=(V',E')$ are isomorphic if there exists a bijection $f: V \to V'$ such that $(u,v) \in E$ if and only if $(f(u), f(v)) \in E'$. In other words, we can relabel vertices of G to be vertices in G', maintaining the corresponding edges in G and G'.

eg:



and



$f(1) = u,\ f(2) = v,\ f(3) = w,\ \ldots\ f(6) = z$

10) Subgraph: A graph $G' = (V', E')$ is said to be the subgraph of $G = (V, E)$ if $V' \subseteq V$ and $E' \subseteq E$. If $V' \subseteq V$, the subgraph of G "induced" by V' is the graph $G' = (V', E')$ where

$$E' = \{ (u,v) \in E : u, v \in V' \}$$

11) To obtain the directed version of an undirected-graph, each undirected edge (u,v) is replaced in the directed version by two directed edges (u,v) and (v,u).

The undirected version of a directed graph, contains the original edges with their directions removed and with self-loops eliminated. If the directed graph contains both edges (u,v) and (v,u), the undirected graph contains just one edge (u,v).

In a directed graph $G = (V, E)$, a neighbour of a vertex u is any vertex that is adjacent to u in the undirected version of G. In an undirected graph u and v are neighbours if they are adjacent.

12) Complete graph: An undirected graph in which every pair of vertices is adjacent.

Bipartite graph: An undirected graph $G = (V, E)$ in which V can be partitioned into 2 sets V_1 and V_2 such that $(u,v) \in E$ implies that either $u \in V_1$ and $v \in V_2$ or $u \in V_2$ and $v \in V_1$. That is all edges go between the sets V_1 and V_2.

Forest: An acyclic, undirected graph.

Free-tree: A connected, acyclic, undirected graph (caug).

Dag: Directed acyclic graph.

Multigraph: A multigraph is like an undirected graph, but it can also have multiple edges between vertices and self loops.

Hypergraph: A hypergraph is like an undirected graph, which has hyperedges. A hyperedge rather than connecting 2 vertices, connects an arbitrary subset of vertices.

Contraction: The contraction of an undirected graph $G = (V, E)$ by an edge $e = (u, v)$ is a graph $G' = (V', E')$ where $V' = \overline{V - \{u, v\} \cup \{x\}}$ and x is a new vertex. The set E' is formed by deleting the edge (u,v) from E and every vertex w incident to u or v, deleting (u, w) or (v, w) and adding the new edge (x, w).

1B) . Inside asymptotic notation V denotes $|V|$ and E denotes $|E|$. Thus, when we say that the running time of an algorithm is $O(VE)$ we mean that the algorithm runs in $O(|V||E|)$ time.